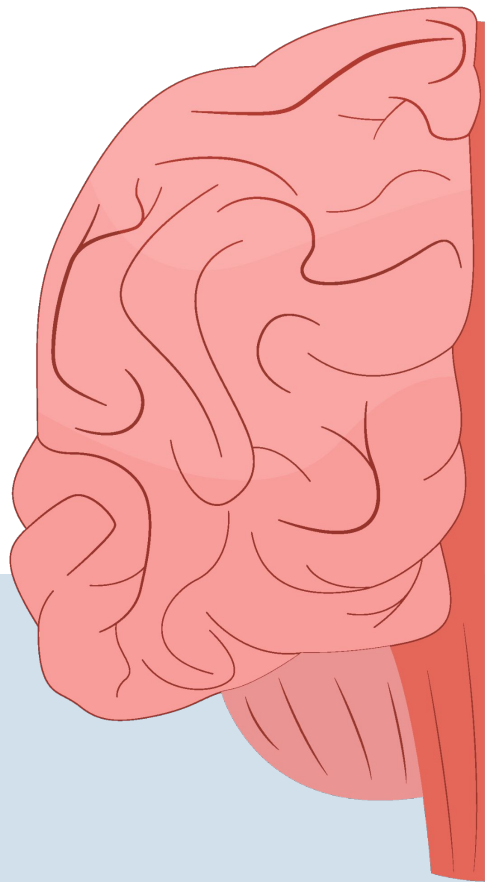
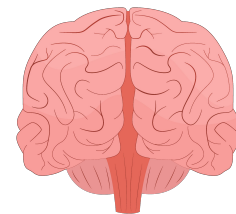




Predicting Mental Health Symptom Severity Using Behavioral and Neuroimaging Data

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Description

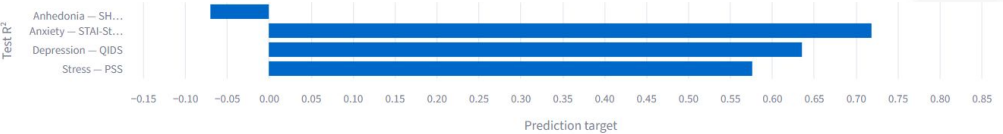
- This project aims to predict continuous symptom severity using behavioral + imaging data through the random forest regression.
 - Metrics: **QIDS** (depression), **STAI-State** (state anxiety), **SHAPS** (anhedonia), and **PSS** (perceived stress).
 - Compares 4 behavioral, imaging, and 4 multimodal models to determine which best predicts symptom severity.
- **Goal:** determine which data modality provides the strongest predictive performance
- **Project Timeline:**
 - Selected multimodal mental health dataset (behavioral + fMRI).
 - Built 28 Random Forest Regression models across four symptom targets.
 - Compared behavioral, imaging, and multimodal feature sets.
 - Discovered and addressed target leakage by removing symptom-scale predictors.
 - Expanded to 36 models with a new non-symptom feature set.
 - Validated that behavioral characteristics remained strong predictors even without overlapping symptom questionnaires.
- **Challenge:** Some behavioral questionnaires measured same mental health symptoms as prediction target → target leakage + performance inflation
 - **Solution:** Added non-symptom behavioral feature set by removing symptom questionnaires from the predictors.
 - The models continued to perform well, showing that personality, cognitive, and lifestyle factors provide meaningful predictive information independent of symptom scales.



Best Model by Target

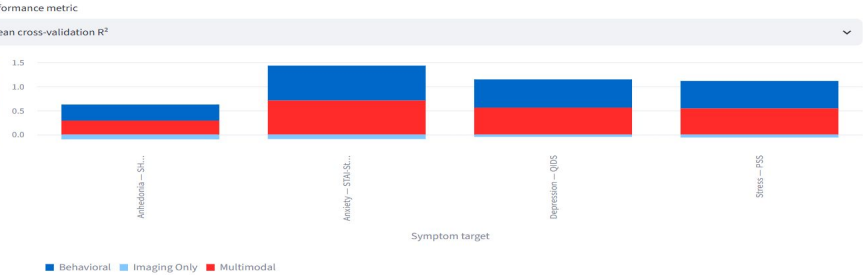
Target	Best Model	Test R ²	CV R ² Mean
Stress — PSS	All Behavioral	0.577	0.572
Depression — QIDS	Relevant Behavioral	0.636	0.588
Anhedonia — SHAPS	All Behavioral	-0.070	0.335
Anxiety — STAI-State	All Behavioral	0.719	0.726

Best Held-Out Test Performance



Behavioral, Imaging, and Multimodal Performance

Select a performance metric to compare the three major sources of predictor data across all four symptom targets.



Higher values indicate better performance. Behavioral models consistently outperformed imaging-only models, and adding imaging generally provided little benefit.

Mental Health Symptom Severity Predictor

Explore how behavioral questionnaires, resting-state fMRI features, and multimodal data perform when predicting mental health symptom severity in the Transdiagnostic Connectome Project dataset.

[Dashboard](#) [Best Models](#) [Interactive Comparison](#) [Key Visualizations](#) [Findings & Outcomes](#)

Project Overview

Prediction Targets	Models Evaluated	Dataset Participants	Model Type
4	36	224	Random Forest

Main Findings

Behavioral-only models performed best

Questionnaire features consistently produced the strongest symptom-severity predictions.

Imaging performance was limited

Resting-state fMRI features alone generally produced weak or negative predictive R² values.

Removing symptom-scale features indicated predictive performance

Non-symptom behavioral characteristics provide valuable information for predicting mental health outcomes.

Interactive Model Comparison

Select a symptom target and a model configuration. The selected model will be shown alongside the other available models for the same target.

Prediction target	Predictor and model configuration
Anxiety — STAI-State	Non-symptom Behavioral

Anxiety — STAI-State: Non-symptom Behavioral

Test R ²	CV R ² Mean	Test MAE	Test RMSE
0.638	0.658	6.638	8.272

Cross-validation R² standard deviation: 0.057

Mental Health Symptom Severity Predictor

WhatsApp

UNiDAYS - Fast, free...

Free Florida (FL) DM...

Symbolab Math Sol...

Welcome - Einthusan

INIT

Class E Knowledge...

GitHub · Build and s...

CCC Contest System

All Bookmarks

Deploy

Project Navigation

Use the tabs at the top of the page to explore:

Project dashboard

Best-performing models

Interactive model comparisons

Findings, limitations, and future work

Prediction Targets

QIDS: Depression

STAI-State: Anxiety

SHAPS: Anhedonia

PSS: Perceived Stress

Mental Health Symptom Severity Predictor

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Main Findings

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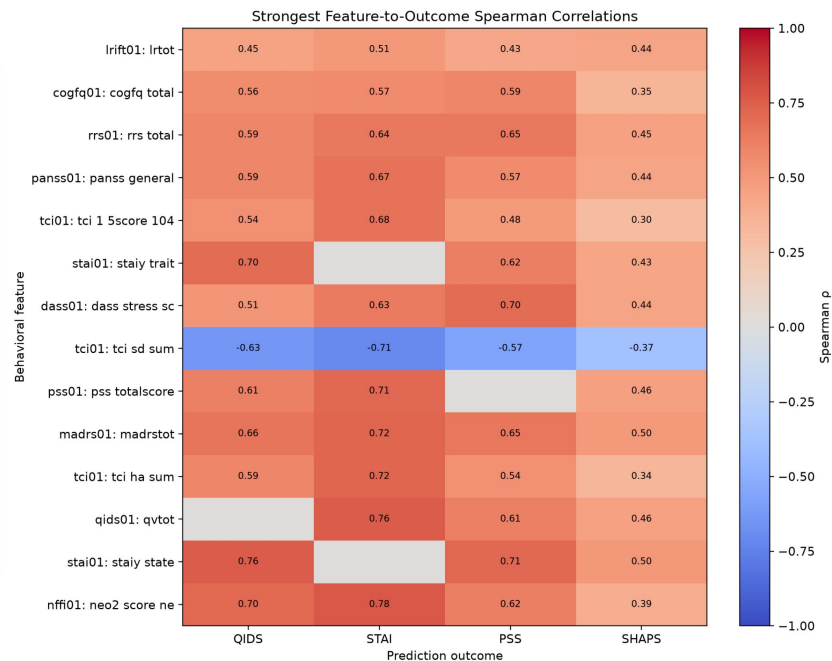
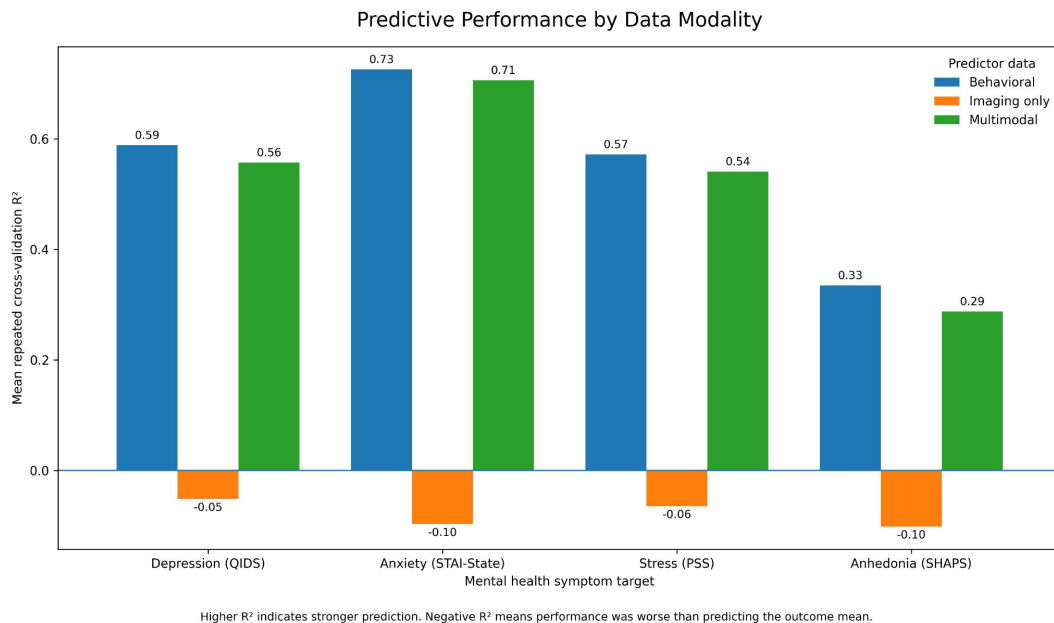
Removing symptom-scale features indicated predictive performance

This application is for educational

Screen Recorder is sharing your screen. Stop sharing Hide

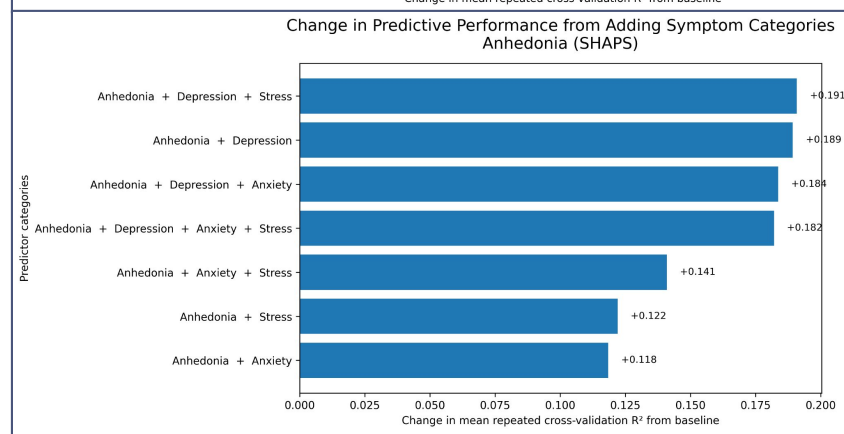
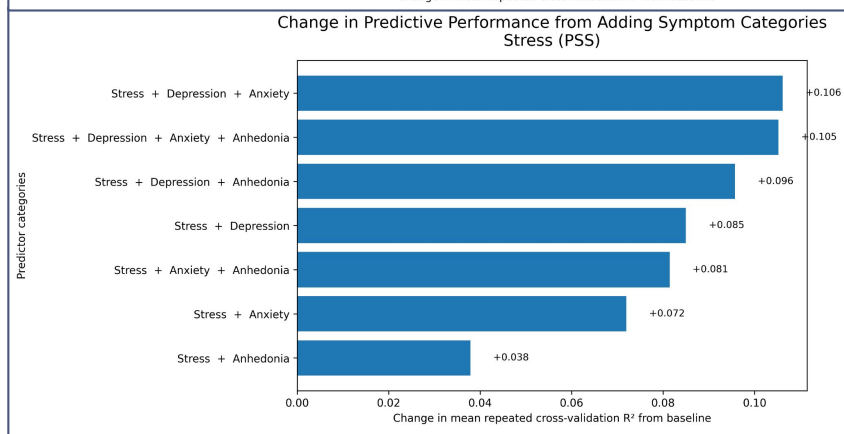
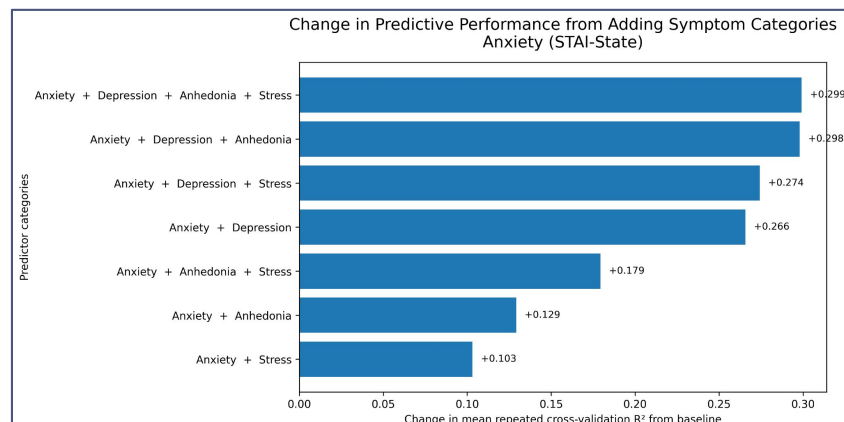
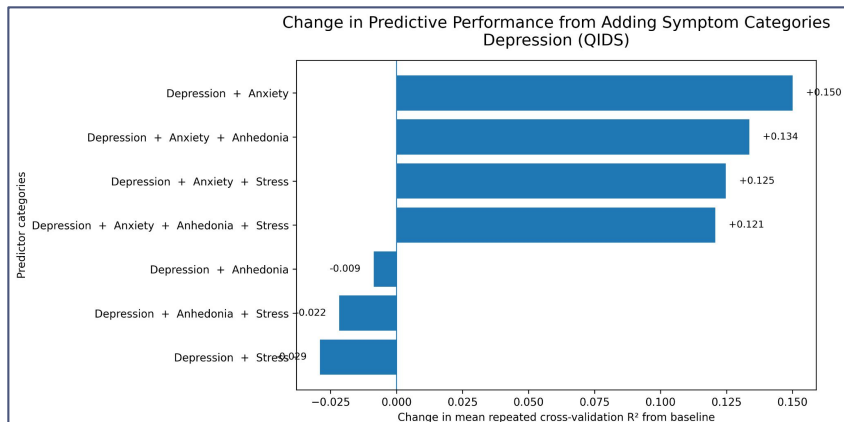
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How Well Did Models Predict Mental Health Symptoms?



Key Takeaway: Behavioral questionnaire data consistently outperformed imaging features, while no behavioral feature showed evidence of excessive correlation ($\rho \geq 0.80$) with its prediction target.

How do cross-category symptom domains influence prediction?



*Positive values indicate improvement over baseline

Key Takeaway: Related symptom domains often improved prediction, but the amount of improvement differed across mental health outcomes.

Algorithm

Supervised Random Forest Regression

- Inputs: behavioral data from questionnaires (all, relevant, category, non-symptom), imaging data from f-MRI scans, and multimodal data (behavioral + imaging).
- These predictors were used for QIDS, STAI-State, PSS, and SHAPS which represent depression, anxiety, stress, and anhedonia respectively.
- Random Forest Regression builds many decision trees and combines their predictions to produce a final result.
 - This helps capture complex relationships between behavioral and imaging variables while reducing overfitting compared to a single decision tree.

Essential Question

How can the process of creating AI/ML solutions amplify or mitigate bias in the case our group is using?

Bias enters at every step of the process. Our choices decide which way it goes.

What amplifies bias

- Only 241 people, all from one study
- Self report scales carry their own bias
- Leaky features can fake a good score

What reduces bias

- Removed symptom scales to stop leakage
- Compared 36 models, not just one
- Shared our weak results and all our code

We cannot remove bias, only make it visible. That is why this stays a research tool and never a diagnosis.

Key Takeaways



Major Findings

- Behavioral features consistently outperformed imaging features
- Multimodal models provided little improvement over behavioral data alone
- Cross-category symptom information improved prediction for some outcomes
 - Removing behavioral symptom questionnaires that could introduce target leakage outperformed imaging features



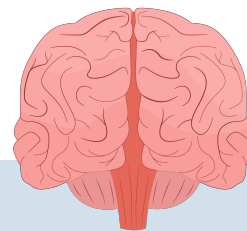
Project Outcomes

- Developed an end-to-end machine learning pipeline
- Compared behavioral, imaging-only, and multimodal models
- Built an interactive Streamlit dashboard for exploring model results



Lessons learned

- Preprocessing and feature engineering greatly affects model performance
- More features do not necessarily produce better predictions
- Thorough validation is essential for trustworthy machine learning models



Positive and Negative Impact

Benefits



Earlier Identification

May help identify individuals at risk for worsening mental health symptoms earlier.



Improved Research

Enables researchers to study relationships between behavioral measures and brain imaging more efficiently.



Clinical Decision Support

Could assist clinicians by providing an additional source of information alongside professional evaluation.



Better Understanding of Mental Health

Cross-category analyses reveal how different symptom domains relate to one another.

Challenges



Not a Diagnostic Tool

Predictions should never replace evaluation by a qualified mental health professional.



Limited Generalizability

Models trained on one research dataset may not perform equally well in different populations.

Imaging Limitations



Resting-state fMRI features alone showed limited predictive value in this study.

Privacy & Ethics



Mental health and neuroimaging data require careful handling to protect participant privacy and confidentiality.

GitHub Link + Next Steps

Limitations

- Small dataset.
- Only Random Forest models were evaluated.
- Resting-state fMRI features may not capture all relevant brain information.
- Single dataset limits generalizability.

Future work

- Test additional machine learning models.
- Improve imaging feature engineering.
- Evaluate other neuroimaging modalities.
- Validate on external datasets.
- Increase sample size.

Citations

- Chopra, S., Cocuzza, C. V., Lawhead, C., Ricard, J. A., Labache, L., Patrick, L. M., Kumar, P., Rubenstein, A., Moses, J., Chen, L., Blankenbaker, C., Gillis, B., Germine, L. T., Harpaz-Rotem, I., Yeo, B. T. T., Baker, J. T., & Holmes, A. J. (2025). *The Transdiagnostic Connectome Project: an open dataset for studying brain-behavior relationships in psychiatry*. Scientific Data, 12, 923. <https://doi.org/10.1038/s41597-025-04895-z>
- Parkes, L., Satterthwaite, T. D., & Bassett, D. S. (2020). *Towards precise resting-state fMRI biomarkers in psychiatry: synthesizing developments in transdiagnostic research, dimensional models of psychopathology, and normative neurodevelopment*. Current Opinion in Neurobiology, 65, 120–128.
- Chopra, S., Cocuzza, C. V., Lawhead, C., et al. (2024). *The Transdiagnostic Connectome Project dataset: a richly phenotyped open resource for advancing the study of brain-behavior relationships in psychiatry*. medRxiv. <https://doi.org/10.1101/2024.06.18.24309054>
- Woo, C.-W., Chang, L. J., Lindquist, M. A., & Wager, T. D. (2017). *Building better biomarkers: Brain models in translational neuroimaging*. Nature Neuroscience, 20, 365–377.
- Bzdok, D., & Meyer-Lindenberg, A. (2018). *Machine learning for precision psychiatry: opportunities and challenges*. Biological Psychiatry: Cognitive Neuroscience and Neuroimaging, 3(3), 223–230.
- Insel, T. R. (2014). *The NIMH Research Domain Criteria (RDoC) Project: precision medicine for psychiatry*. American Journal of Psychiatry, 171(4), 395–397.
- Yuwon Kim. (2025). *Beyond Missing Data: A Multimodal Approach Using VR-EEG-MRI (VEEM) Biomarkers for Detecting MCI*. In Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25). Association for Computing Machinery, New York, NY, USA, Article 895, 1–7. <https://doi.org/10.1145/3706599.3719284>
- Jinxia Wang, Liang Qiao, Haibin Lv, and Zhihan Lv. 2022. *Deep Transfer Learning-Based Multi-Modal Digital Twins for Enhancement and Diagnostic Analysis of Brain MRI Image*. IEEE/ACM Trans. Comput. Biol. Bioinformatics 20, 4 (July-Aug. 2023), 2407–2419. <https://doi.org/10.1109/TCBB.2022.3168189>